# Final

July 27, 2025

# 1 Identifying Low-Risk Aircraft for New Aviation Division Launch

## 1.1 Project Overview

- This project is a strategic initiative to support the company's expansion into the aviation industry with data-driven insights to inform initial aircraft acquisition decisions.
- By thoroughly analyzing historical aviation incident data, this project will quantify key risk factors such as accident frequency, damage severity, and human injury rates for various aircraft makes and models.
- The ultimate goal is to identify and recommend aircraft that exhibit the lowest overall risk profiles, thereby enabling the head of the new division to make strategically sound purchases that minimize financial liabilities, operational disruptions, and safety concerns, ensuring a secure and successful entry into the aviation industry.

## 1.2 Business Understanding

### 1.2.1 Project Purpose

This project is a strategic initiative to support the company's expansion into the aviation industry. The primary purpose is to conduct a comprehensive analysis of aircraft risk profiles to identify and recommend the most suitable low-risk aircraft for the new aviation division's initial fleet. This analysis will provide the necessary data and insights to mitigate the significant risks associated with aircraft acquisition and operation, thereby ensuring a stable and successful launch of the new business venture.

### 1.2.2 Problem Statement

The company is entering the aviation sector with a limited understanding of the inherent risks, which include, but are not limited to, accident rates, operational complexity, aircraft damage potential, and the human cost of incidents. A lack of data-driven insights in this area could lead to sub-optimal purchasing decisions, resulting in financial losses, operational inefficiencies, and significant safety liabilities. The problem is to transform raw aviation incident data into actionable intelligence that quantifies these risks and guides the selection of a low-risk, high-reliability aircraft portfolio.

### 1.2.3 Project Scope

This project will focus on the following key areas: Data Analysis: Utilizing the provided "Aviation_Data.csv" dataset, the project will analyze historical aviation incidents to assess risk factors.

- Risk Factor Quantification: Key risk factors such as accident frequency, aircraft damage severity, and human injury rates will be quantified and benchmarked across different aircraft makes and models.
- Aircraft Filtering: The analysis will be filtered to focus on aircraft types relevant to commercial and private enterprises, excluding categories like amateur-built.
- Risk Profile Generation: A comprehensive risk profile will be developed for each significant aircraft make and model, considering accident severity, operational context (e.g., phase of flight, weather conditions), and human impact.
- Recommendation & Actionable Insights: The project will culminate in a set of clear, actionable recommendations for the head of the new aviation division, identifying specific aircraft makes and models that represent the lowest overall risk for initial acquisition. The report will explain the rationale behind these recommendations, highlighting key risk mitigation strategies.

#### 1.2.4 Key Business Questions to Address:

1. What are the primary risk factors associated with aircraft ownership and operation (e.g., accident rates,

   operational complexity, human impact)?
2. How can these risk factors be quantified or assessed for different aircraft types?
3. Which specific aircraft models exhibit the lowest overall risk profile based on a comprehensive evaluation of these factors?
4. What are the key characteristics of these low-risk aircraft that make them suitable for the company's initial ventures?
5. What actionable insights and recommendations can be provided to the head of the aviation division to guide their aircraft purchasing

   decisions?

### 1.3 Data Understanding

The data sources for this analysis will be pulled from the National Transportation Safety Board that includes aviation accident data from 1962 to 2023 about civil aviation accidents and selected incidents in the United States and international waters. The data is contained in a csv file "Aviation_Data.csv"

#### 1.3.1 Loading the Data with Pandas

In the cell below, we:

- Import and alias pandas as pd
- Import and alias numpy as np
- Import and alias seaborn as sns
- Import and alias matplotlib.pyplot as plt
- Set Matplotlib visualizations to display inline in the notebook

[1]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
```

[2]:
```
df = pd.read_csv('Aviation_Data.csv', low_memory=False )
df.head()
```

[2]:

| | Event.Id | Investigation.Type | Accident.Number | Event.Date | Location | Country | Latitude | Longitude | Airport.Code | Airport.Name | ... | Purpose.of.flight | Air.carrier | Total.Fatal.Injuries | Total.Serious.Injuries | Total.Minor.Injuries | Total.Uninjured | Weather.Condition | Broad.phase.of.flight | Report.Status | Publication.Date |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 20001218X45444 | Accident | SEA87LA080 | 1948-10-24 | MOOSE CREEK, ID | United States | NaN | NaN | NaN | NaN | ... | Personal | NaN | 2.0 | 0.0 | 0.0 | 0.0 | UNK | Cruise | Probable Cause | NaN |
| 1 | 20001218X45447 | Accident | LAX94LA336 | 1962-07-19 | BRIDGEPORT, CA | United States | NaN | NaN | NaN | NaN | ... | Personal | NaN | 4.0 | 0.0 | 0.0 | 0.0 | UNK | Unknown | Probable Cause | 19-09-1996 |
| 2 | 20061025X01555 | Accident | NYC07LA005 | 1974-08-30 | Saltville, VA | United States | 36.922223 | -81.878056 | NaN | NaN | ... | Personal | NaN | 3.0 | NaN | NaN | NaN | IMC | Cruise | Probable Cause | 26-02-2007 |
| 3 | 20001218X45448 | Accident | LAX96LA321 | 1977-06-19 | EUREKA, CA | United States | NaN | NaN | NaN | NaN | ... | Personal | NaN | 2.0 | 0.0 | 0.0 | 0.0 | IMC | Cruise | Probable Cause | 12-09-2000 |
| 4 | 20041105X01764 | Accident | CHI79FA064 | 1979-08-02 | Canton, OH | United States | NaN | NaN | NaN | NaN | ... | Personal | NaN | 1.0 | 2.0 | NaN | 0.0 | VMC | Approach | Probable Cause | 16-04-1980 |

[5 rows x 31 columns]

### 1.3.2 Famializing with the Data

- Understanding the dimensionality of the dataset
- Investigating what type of data it contains, and the data types used to store it
- Discovering how missing values are encoded, and how many there are
- Getting a feel for what information it does and doesn't contain

[3]:
```
# dimensionality of the dataset
df.shape
```

[3]:
```
(90348, 31)
```

[4]:
```
# Investigating what type of data it contains, and the data types used to store␣
↪it
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 90348 entries, 0 to 90347
Data columns (total 31 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Event.Id                88889 non-null  object
 1   Investigation.Type      90348 non-null  object
 2   Accident.Number         88889 non-null  object
 3   Event.Date              88889 non-null  object
 4   Location                88837 non-null  object
 5   Country                 88663 non-null  object
 6   Latitude                34382 non-null  object
 7   Longitude               34373 non-null  object
 8   Airport.Code            50132 non-null  object
 9   Airport.Name            52704 non-null  object
 10  Injury.Severity         87889 non-null  object
 11  Aircraft.damage         85695 non-null  object
 12  Aircraft.Category       32287 non-null  object
 13  Registration.Number     87507 non-null  object
 14  Make                    88826 non-null  object
 15  Model                   88797 non-null  object
 16  Amateur.Built           88787 non-null  object
 17  Number.of.Engines       82805 non-null  float64
 18  Engine.Type             81793 non-null  object
 19  FAR.Description         32023 non-null  object
 20  Schedule                12582 non-null  object
 21  Purpose.of.flight       82697 non-null  object
 22  Air.carrier             16648 non-null  object
 23  Total.Fatal.Injuries    77488 non-null  float64
 24  Total.Serious.Injuries  76379 non-null  float64
 25  Total.Minor.Injuries    76956 non-null  float64
 26  Total.Uninjured         82977 non-null  float64
 27  Weather.Condition       84397 non-null  object
```

```
 28  Broad.phase.of.flight   61724 non-null  object
 29  Report.Status           82505 non-null  object
 30  Publication.Date        73659 non-null  object
dtypes: float64(5), object(26)
memory usage: 21.4+ MB
```

From the information above we can deduce that: most of the items are objects and only 5 out of the 30 columns are floats. And Event.Date and Publication.Date are objects instead of datetime.

[5]:
```
# Discovering how missing values are encoded, and how many there are
df.isnull().sum()
```

[5]:
```
Event.Id                 1459
Investigation.Type          0
Accident.Number          1459
Event.Date               1459
Location                 1511
Country                  1685
Latitude                55966
Longitude               55975
Airport.Code            40216
Airport.Name            37644
Injury.Severity          2459
Aircraft.damage          4653
Aircraft.Category       58061
Registration.Number      2841
Make                     1522
Model                    1551
Amateur.Built            1561
Number.of.Engines        7543
Engine.Type              8555
FAR.Description         58325
Schedule                77766
Purpose.of.flight        7651
Air.carrier             73700
Total.Fatal.Injuries    12860
Total.Serious.Injuries  13969
Total.Minor.Injuries    13392
Total.Uninjured          7371
Weather.Condition        5951
Broad.phase.of.flight   28624
Report.Status            7843
Publication.Date        16689
dtype: int64
```

[6]:
```
df['Amateur.Built'].unique()
```

[6]:
```
array(['No', 'Yes', nan], dtype=object)
```

[7]:
```
df['Aircraft.Category'].unique()
```

[7]:
```
array([nan, 'Airplane', 'Helicopter', 'Glider', 'Balloon', 'Gyrocraft',
       'Ultralight', 'Unknown', 'Blimp', 'Powered-Lift', 'Weight-Shift',
       'Powered Parachute', 'Rocket', 'WSFT', 'UNK', 'ULTR'], dtype=object)
```

[8]:
```
# Getting a feel for what information it does and doesn't contain
df.columns
```

[8]:
```
Index(['Event.Id', 'Investigation.Type', 'Accident.Number', 'Event.Date',
       'Location', 'Country', 'Latitude', 'Longitude', 'Airport.Code',
       'Airport.Name', 'Injury.Severity', 'Aircraft.damage',
       'Aircraft.Category', 'Registration.Number', 'Make', 'Model',
       'Amateur.Built', 'Number.of.Engines', 'Engine.Type', 'FAR.Description',
       'Schedule', 'Purpose.of.flight', 'Air.carrier', 'Total.Fatal.Injuries',
       'Total.Serious.Injuries', 'Total.Minor.Injuries', 'Total.Uninjured',
       'Weather.Condition', 'Broad.phase.of.flight', 'Report.Status',
       'Publication.Date'],
      dtype='object')
```

[9]:
```
df['Aircraft.damage'].unique()
```

[9]:
```
array(['Destroyed', 'Substantial', 'Minor', nan, 'Unknown'], dtype=object)
```

[10]:
```
df['Aircraft.Category'].unique()
```

[10]:
```
array([nan, 'Airplane', 'Helicopter', 'Glider', 'Balloon', 'Gyrocraft',
       'Ultralight', 'Unknown', 'Blimp', 'Powered-Lift', 'Weight-Shift',
       'Powered Parachute', 'Rocket', 'WSFT', 'UNK', 'ULTR'], dtype=object)
```

This dataset primarily contains incident data which is excellent for analyzing accident rates, injury severity, and aircraft damage, which are critical risk factors. Below is a sample of columns and the description of what they contain.

- **Aircraft.damage**: This is crucial. It directly indicates the severity of an incident. "Destroyed" or "Substantial" damage implies higher risk and potentially higher costs.
- **Injury.Severity**: This column is paramount as it reflects the human cost of an accident. Total Fatal, Serious, and Minor Injuries, as well as Uninjured, provide a comprehensive picture of the impact on human life, which is a key risk factor for any company.
- **Make**: This identifies the manufacturer of the aircraft. Different manufacturers have different safety records, design philosophies, and support networks.
- **Model**: This is even more specific than 'Make'. Different models within the same manufacturer can have vastly different risk profiles due to design, age, or operational history.
- **Aircraft.Category**: This classifies the type of aircraft (e.g., "Airplane," "Rotorcraft," "Glider"). Different categories have inherent risk differences and different operational requirements.

- **Total.Fatal.Injuries, Total.Serious.Injuries, Total.Minor.Injuries, Total.Uninjured**: These provide granular details on the human impact of each event, allowing for a more nuanced risk assessment beyond just "Injury.Severity".
- **Accident.Number (or Event.Id)**: This columns serves as a unique identifier for each event. Can be used to group and count incidents by aircraft make/model to calculate accident rates.
- **Event.Date**: Useful for analyzing trends over time.
- **Purpose.of.flight**: The purpose of the flight (e.g., "Commercial," "Personal," "Instructional") can significantly influence accident types and frequencies.
- **Weather.Condition**: Adverse weather is a known factor in accidents. Analyzing this can help understand if certain aircraft types are more susceptible to weather-related incidents.
- **Broad.phase.of.flight**: This column provides context for when accidents occur, which can highlight specific vulnerabilities of an aircraft during different operational phases.

### 1.4 Data Cleaning and Filtering

[11]:
```
# Cleaning the column names
df.columns = [col.replace(".", "_") for col in df.columns]
df.columns
```

[11]:
```
Index(['Event_Id', 'Investigation_Type', 'Accident_Number', 'Event_Date',
       'Location', 'Country', 'Latitude', 'Longitude', 'Airport_Code',
       'Airport_Name', 'Injury_Severity', 'Aircraft_damage',
       'Aircraft_Category', 'Registration_Number', 'Make', 'Model',
       'Amateur_Built', 'Number_of_Engines', 'Engine_Type', 'FAR_Description',
       'Schedule', 'Purpose_of_flight', 'Air_carrier', 'Total_Fatal_Injuries',
       'Total_Serious_Injuries', 'Total_Minor_Injuries', 'Total_Uninjured',
       'Weather_Condition', 'Broad_phase_of_flight', 'Report_Status',
       'Publication_Date'],
      dtype='object')
```

[12]:
```
# checking for duplicate rows in the data
df.duplicated().value_counts()
```

[12]:
```
False    88958
True      1390
Name: count, dtype: int64
```

[13]:
```
# Grouping the duplicates and confirming the data they contain
df[df.duplicated(keep=False)].sort_values(by='Event_Id')
```

[13]:

| | Event_Id | Investigation_Type | Accident_Number | Event_Date | Location | Country | \ |
|---|---|---|---|---|---|---|---|
| 64030 | NaN | | 25-09-2020 | NaN | NaN | NaN | NaN |
| 64050 | NaN | | 25-09-2020 | NaN | NaN | NaN | NaN |
| 64052 | NaN | | 25-09-2020 | NaN | NaN | NaN | NaN |
| 64388 | NaN | | 25-09-2020 | NaN | NaN | NaN | NaN |

```
64541      NaN         25-09-2020             NaN        NaN      NaN     NaN
...        ...                ...             ...        ...      ...     ...
90004      NaN         15-12-2022             NaN        NaN      NaN     NaN
90010      NaN         15-12-2022             NaN        NaN      NaN     NaN
90031      NaN         15-12-2022             NaN        NaN      NaN     NaN
90090      NaN         20-12-2022             NaN        NaN      NaN     NaN
90097      NaN         20-12-2022             NaN        NaN      NaN     NaN

      Latitude Longitude Airport_Code Airport_Name  ... Purpose_of_flight  \
64030      NaN       NaN          NaN          NaN  ...                NaN
64050      NaN       NaN          NaN          NaN  ...                NaN
64052      NaN       NaN          NaN          NaN  ...                NaN
64388      NaN       NaN          NaN          NaN  ...                NaN
64541      NaN       NaN          NaN          NaN  ...                NaN
...        ...       ...          ...          ...  ...                ...
90004      NaN       NaN          NaN          NaN  ...                NaN
90010      NaN       NaN          NaN          NaN  ...                NaN
90031      NaN       NaN          NaN          NaN  ...                NaN
90090      NaN       NaN          NaN          NaN  ...                NaN
90097      NaN       NaN          NaN          NaN  ...                NaN

      Air_carrier Total_Fatal_Injuries Total_Serious_Injuries  \
64030         NaN                  NaN                    NaN
64050         NaN                  NaN                    NaN
64052         NaN                  NaN                    NaN
64388         NaN                  NaN                    NaN
64541         NaN                  NaN                    NaN
...           ...                  ...                    ...
90004         NaN                  NaN                    NaN
90010         NaN                  NaN                    NaN
90031         NaN                  NaN                    NaN
90090         NaN                  NaN                    NaN
90097         NaN                  NaN                    NaN

      Total_Minor_Injuries Total_Uninjured Weather_Condition  \
64030                  NaN             NaN               NaN
64050                  NaN             NaN               NaN
64052                  NaN             NaN               NaN
64388                  NaN             NaN               NaN
64541                  NaN             NaN               NaN
...                    ...             ...               ...
90004                  NaN             NaN               NaN
90010                  NaN             NaN               NaN
90031                  NaN             NaN               NaN
90090                  NaN             NaN               NaN
90097                  NaN             NaN               NaN
```

```
       Broad_phase_of_flight Report_Status Publication_Date
64030                    NaN           NaN              NaN
64050                    NaN           NaN              NaN
64052                    NaN           NaN              NaN
64388                    NaN           NaN              NaN
64541                    NaN           NaN              NaN
...                      ...           ...              ...
90004                    NaN           NaN              NaN
90010                    NaN           NaN              NaN
90031                    NaN           NaN              NaN
90090                    NaN           NaN              NaN
90097                    NaN           NaN              NaN

[1447 rows x 31 columns]
```

[14]:
```
# Dropping the rows of duplicates
df.drop_duplicates(inplace=True)
```

[15]:
```
# Convert Event_Date to datetime
df['Event_Date'] = pd.to_datetime(df['Event_Date'], errors='coerce')
```

[16]:
```
# Filtering out only airplanes and ones that are not Amateur Built.
# The business problem mentions "airplanes for commercial and private␣
 ↪enterprises"
df_filtered = df[df['Aircraft_Category'] == 'Airplane'].copy()
df_filtered = df_filtered[df_filtered['Amateur_Built']!= "Yes"]
df_filtered
```

[16]:
```
             Event_Id Investigation_Type Accident_Number Event_Date  \
5      20170710X52551           Accident      NYC79AA106 1979-09-17
7      20020909X01562           Accident      SEA82DA022 1982-01-01
8      20020909X01561           Accident      NYC82DA015 1982-01-01
12     20020917X02148           Accident      FTW82FRJ07 1982-01-02
13     20020917X02134           Accident      FTW82FRA14 1982-01-02
...               ...                ...             ...        ...
90328  20221213106455           Accident      WPR23LA065 2022-12-13
90332  20221215106463           Accident      ERA23LA090 2022-12-14
90335  20221219106475           Accident      WPR23LA069 2022-12-15
90336  20221219106470           Accident      ERA23LA091 2022-12-16
90345  20221227106497           Accident      WPR23LA075 2022-12-26

                Location        Country   Latitude   Longitude Airport_Code  \
5            BOSTON, MA  United States  42.445277  -70.758333          NaN
7          PULLMAN, WA   United States        NaN         NaN          NaN
8      EAST HANOVER, NJ  United States        NaN         NaN          N58
12           HOMER, LA   United States        NaN         NaN          NaN
13          HEARNE, TX   United States        NaN         NaN          T72
```

```
...                 ...            ...        ...         ...          ...
90328     Lewistown, MT  United States    047257N    0109280W         KLWT
90332      San Juan, PR  United States    182724N    0066554W          SIG
90335       Wichita, KS  United States    373829N    0972635W          ICT
90336   Brooksville, FL  United States    282825N    0822719W          BKV
90345        Payson, AZ  United States    341525N    1112021W          PAN

                        Airport_Name  ... Purpose_of_flight  \
5                                NaN  ...               NaN
7                  BLACKBURN AG STRIP  ...          Personal
8                            HANOVER  ...          Business
12                               NaN  ...          Personal
13                   HEARNE MUNICIPAL  ...          Personal
...                              ...  ...               ...
90328      Lewiston Municipal Airport  ...               NaN
90332   FERNANDO LUIS RIBAS DOMINICCI  ...          Personal
90335  WICHITA DWIGHT D EISENHOWER NT  ...               NaN
90336      BROOKSVILLE-TAMPA BAY RGNL  ...          Personal
90345                          PAYSON  ...          Personal

                       Air_carrier Total_Fatal_Injuries  \
5                       Air Canada                  NaN
7                              NaN                  0.0
8                              NaN                  0.0
12                             NaN                  0.0
13                             NaN                  1.0
...                            ...                  ...
90328                          NaN                  0.0
90332  SKY WEST AVIATION INC TRUSTEE                0.0
90335                          NaN                  0.0
90336               GERBER RICHARD E                0.0
90345                          NaN                  0.0

      Total_Serious_Injuries Total_Minor_Injuries Total_Uninjured  \
5                        NaN                 1.0            44.0
7                        0.0                 0.0             2.0
8                        0.0                 0.0             2.0
12                       0.0                 1.0             0.0
13                       0.0                 0.0             0.0
...                      ...                 ...             ...
90328                    0.0                 0.0             1.0
90332                    0.0                 0.0             1.0
90335                    0.0                 0.0             1.0
90336                    1.0                 0.0             0.0
90345                    0.0                 0.0             1.0

      Weather_Condition  Broad_phase_of_flight   Report_Status  \
```

```
5                   VMC                  Climb  Probable Cause
7                   VMC                Takeoff  Probable Cause
8                   IMC                Landing  Probable Cause
12                  IMC                 Cruise  Probable Cause
13                  IMC                Takeoff  Probable Cause
...                 ...                    ...             ...
90328               NaN                    NaN             NaN
90332               VMC                    NaN             NaN
90335               NaN                    NaN             NaN
90336               VMC                    NaN             NaN
90345               VMC                    NaN             NaN

      Publication_Date
5           19-09-2017
7           01-01-1982
8           01-01-1982
12          02-01-1983
13          02-01-1983
...                ...
90328       14-12-2022
90332       27-12-2022
90335       19-12-2022
90336       23-12-2022
90345       27-12-2022

[24434 rows x 31 columns]
```

[17]:
```
# Asserting that we only have airplanes in category and there are no Amateur␣
 ↪built
df_filtered['Amateur_Built'].value_counts()
df_filtered['Aircraft_Category'].value_counts()
print(f"{df_filtered['Amateur_Built'].value_counts()} and␣
 ↪{df_filtered['Aircraft_Category'].value_counts()}")
```

```
Amateur_Built
No    24417
Name: count, dtype: int64 and Aircraft_Category
Airplane    24434
Name: count, dtype: int64
```

**Further cleaning on key columns** Filling missing values in injury or damage columns is essential before performing calculations on them. For categorical data, some of them already have unknown as a unique variable, makes sense to fill the na in the categorical columns with 'Unknown' to prevent errors.

[18]:
```
# To find the missing values of the df_filtered
df_filtered.isna().sum()
```

[18]:
```
Event_Id                    0
Investigation_Type          0
Accident_Number             0
Event_Date                  0
Location                    7
Country                     7
Latitude                 5259
Longitude                5265
Airport_Code             8948
Airport_Name             8449
Injury_Severity           813
Aircraft_damage          1274
Aircraft_Category           0
Registration_Number       216
Make                        3
Model                      18
Amateur_Built              17
Number_of_Engines        2562
Engine_Type              3967
FAR_Description           480
Schedule                21486
Purpose_of_flight        3695
Air_carrier             14078
Total_Fatal_Injuries     2779
Total_Serious_Injuries   2862
Total_Minor_Injuries     2578
Total_Uninjured           729
Weather_Condition        2990
Broad_phase_of_flight   18635
Report_Status            4663
Publication_Date         2014
dtype: int64
```

[19]:
```
df_filtered['Total_Fatal_Injuries'].describe()
```

[19]:
```
count    21655.000000
mean         0.692912
std          6.304978
min          0.000000
25%          0.000000
50%          0.000000
75%          0.000000
max        295.000000
Name: Total_Fatal_Injuries, dtype: float64
```

[20]:
```
df_filtered['Total_Serious_Injuries'].describe()
```

[20]:
```
count    21572.000000
mean         0.304144
std          2.222139
min          0.000000
25%          0.000000
50%          0.000000
75%          0.000000
max        161.000000
Name: Total_Serious_Injuries, dtype: float64
```

[21]:
```
# Fill missing injury values with 0 it is a sensible choice since the median/50%␣
 ↪for the three is 0.
injury_cols = ['Total_Fatal_Injuries', 'Total_Serious_Injuries',␣
 ↪'Total_Minor_Injuries', 'Total_Uninjured']
for col in injury_cols:
    df_filtered[col] = df_filtered[col].fillna(0).astype(int)
```

[22]:
```
# filling the missing values of the categorical columns that we may need for our␣
 ↪analysis  with 'Unknown'
aircraft_related_cols = ['Injury_Severity', 'Aircraft_damage', 'Make', 'Model',␣
 ↪'Amateur_Built',
       'Weather_Condition', 'Broad_phase_of_flight']
for cols in aircraft_related_cols:
    df_filtered[cols] = df_filtered[cols].fillna('Unknown')
```

[23]:
```
#Investigating the top 20 value counts of 'Make'.
df_filtered['Make'].value_counts().head(20)
```

[23]:
```
Make
CESSNA                4867
Cessna                3578
PIPER                 2803
Piper                 1901
BOEING                1037
BEECH                 1018
Beech                  667
Boeing                 278
MOONEY                 238
CIRRUS DESIGN CORP     218
AIR TRACTOR INC        217
AIRBUS                 215
Mooney                 179
Grumman                172
BELLANCA               158
AERONCA                149
MAULE                  144
Air Tractor            135
```

```
EMBRAER                 123
Bellanca                123
Name: count, dtype: int64
```

From the above we can see that CESSNA and Cessna, PIPER AND Piper, BEECH and Beech are the same 'Make' with different cases. and others that belong to the same make but encoded differently. Below we will deal with a few of them.

[24]:
```
# normalizing the names of the Make
df_filtered['Make'] = df_filtered['Make'].str.title()
df_filtered['Make'].value_counts().head(10)
```

[24]:
```
Make
Cessna      8445
Piper       4704
Beech       1685
Boeing      1315
Mooney       417
Bellanca     281
Grumman      249
Airbus       244
Maule        232
Aeronca      227
Name: count, dtype: int64
```

[25]:
```
df_filtered['Make'] = df_filtered['Make'].apply(lambda x: 'Airbus' if 'Airbus'␣
 ↪in str(x) else x)
df_filtered['Make'] = df_filtered['Make'].apply(lambda x: 'Aeronca' if 'Aeronca'␣
 ↪in str(x) else x)
df_filtered['Make'] = df_filtered['Make'].apply(lambda x: 'Cessna' if 'Cessna'␣
 ↪in str(x) else x)
df_filtered['Make'] = df_filtered['Make'].apply(lambda x: 'Aviat' if 'Aviat' in␣
 ↪str(x) else x)
df_filtered['Make'] = df_filtered['Make'].apply(lambda x: 'Cirrus' if 'Cirrus'␣
 ↪in str(x) else x)
df_filtered['Make'] = df_filtered['Make'].apply(lambda x: 'Air Tractor' if 'Air␣
 ↪Tractor' in str(x) else x)
```

[26]:
```
df_filtered['Make'] = df_filtered['Make'].replace('Dehavilland', 'De Havilland')
```

[27]:
```
# To clean all entries with Fatal and group them all together
df_filtered['Injury_Severity'] = df_filtered['Injury_Severity'].apply(lambda x:␣
 ↪'Fatal' if 'Fatal' in str(x) else x)
```

[28]:
```
# Combining/grouping Unknown and Unavailable as one group.
df_filtered['Injury_Severity'] = df_filtered['Injury_Severity'].
 ↪replace('Unavailable', 'Unknown')
```

```
df_filtered['Injury_Severity'].value_counts()
```

[28]:
```
Injury_Severity
Fatal       23112
Unknown       830
Incident      238
Minor         146
Serious       108
Name: count, dtype: int64
```

[29]:
```
df_filtered['Weather_Condition'] = df_filtered['Weather_Condition'].str.upper()
```

[30]:
```
df_filtered['Weather_Condition'].value_counts()
```

[30]:
```
Weather_Condition
VMC        19715
UNKNOWN     2990
IMC         1365
UNK          364
Name: count, dtype: int64
```

[31]:
```
df_filtered['Weather_Condition'] = df_filtered['Weather_Condition'].
 ↪replace('UNKNOWN', 'UNK')
```

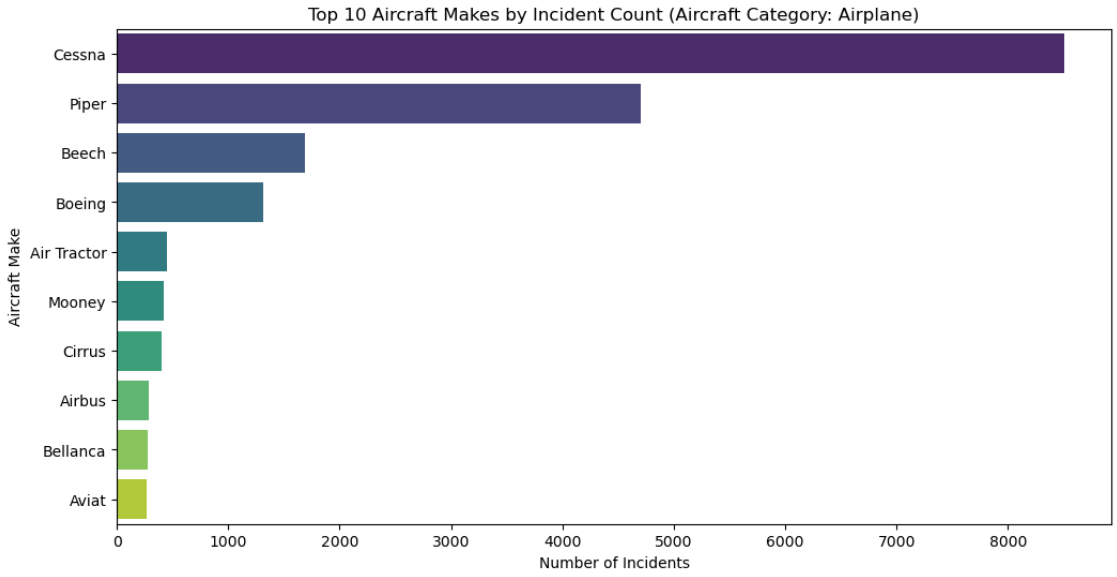
## 1.5 Analyzing Key Risk Factors

This is fundamental. In order to know which aircraft types are involved in more accidents and incidents. Analyzing Key Risk Factors (as per dataset capabilities):

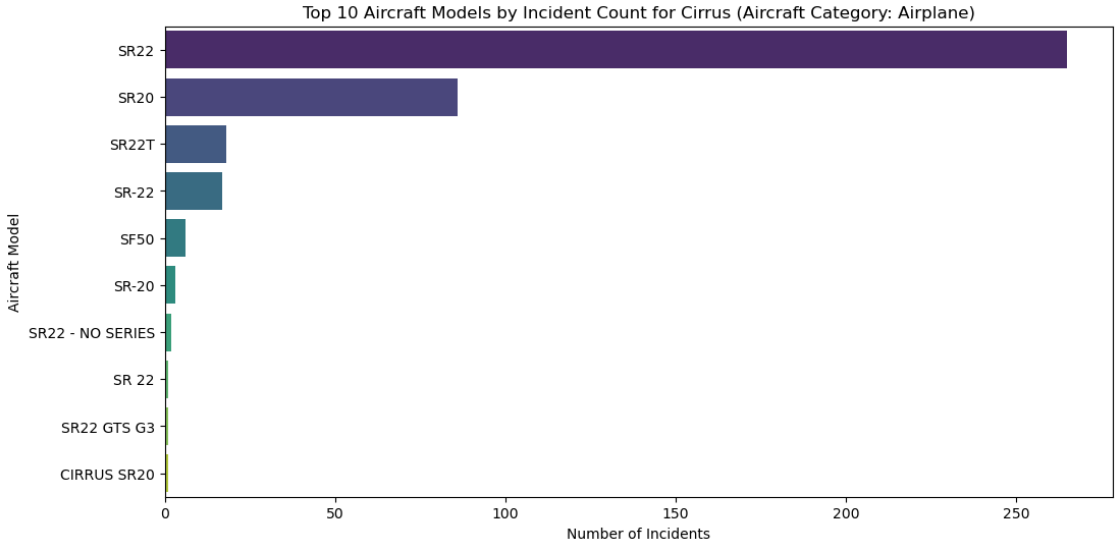
### 1.5.1 Incident Rates by Make and Model

- This directly addresses "accident rates." By counting Make and Model occurrences.
- This is a raw count, not a rate per flight hour or per aircraft in service.
- A model might appear frequently simply because many of them exist.
- We will be focussing primarily the top 10 make and model.

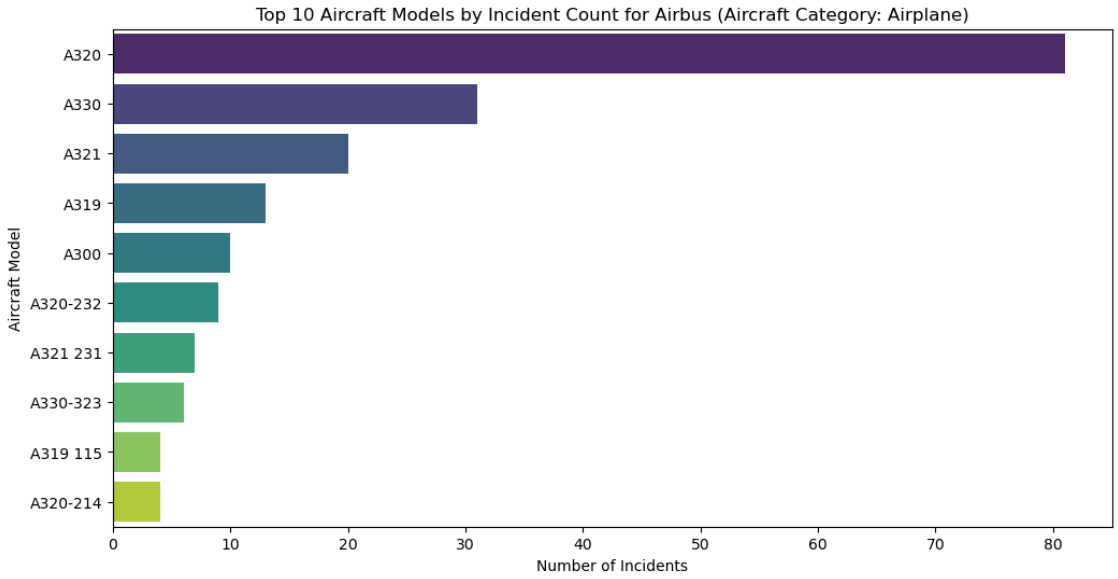
[32]:
```
# The Top 10 Aircraft Makes by Incident Count
make_incidents = df_filtered['Make'].value_counts().head(10)

plt.figure(figsize=(12, 6))
sns.barplot(x=make_incidents.values, y=make_incidents.index, hue=make_incidents.
 ↪index, palette='viridis', legend=False)
plt.title('Top 10 Aircraft Makes by Incident Count (Aircraft Category:␣
 ↪Airplane)')
plt.xlabel('Number of Incidents')
plt.ylabel('Aircraft Make')
plt.show()
```



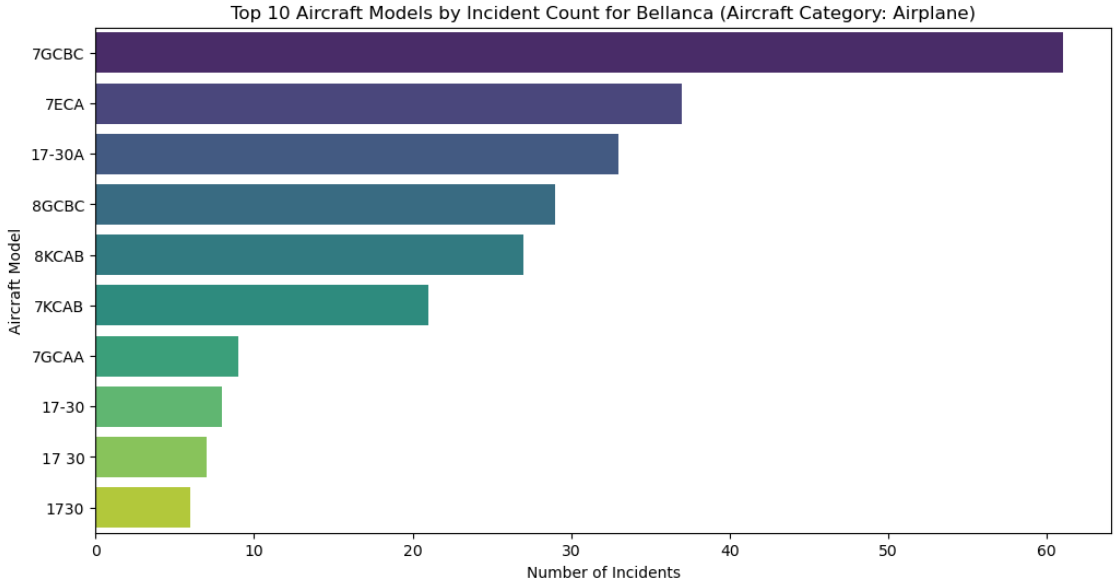

[33]:
```
# Top 10 Aircraft Models by Incident Count which we will filter by Make and␣
 ↪focus on the lowest count incidents of make
# Example: Let's look at models for the top Make(10) (e.g., Cirrus,␣
 ↪Airbus,Bellanca and Aviat)
 # Get the make through indexing the top 10 most frequent make
top_make_6 = make_incidents.index[6] # Cirrus
top_make_7 = make_incidents.index[7] # Airbus
top_make_8 =make_incidents.index[8] # Bellanca
top_make_9 = make_incidents.index[9] # Aviat

model_incidents = df_filtered[df_filtered['Make'] == top_make_6]['Model'].
 ↪value_counts().head(10)

plt.figure(figsize=(12, 6))
sns.barplot( x=model_incidents.values, y=model_incidents.index,␣
 ↪hue=make_incidents.index, palette='viridis',legend=False )
plt.title(f'Top 10 Aircraft Models by Incident Count for {top_make_6} (Aircraft␣
 ↪Category: Airplane)')
plt.xlabel('Number of Incidents')
plt.ylabel('Aircraft Model')
plt.show()
```



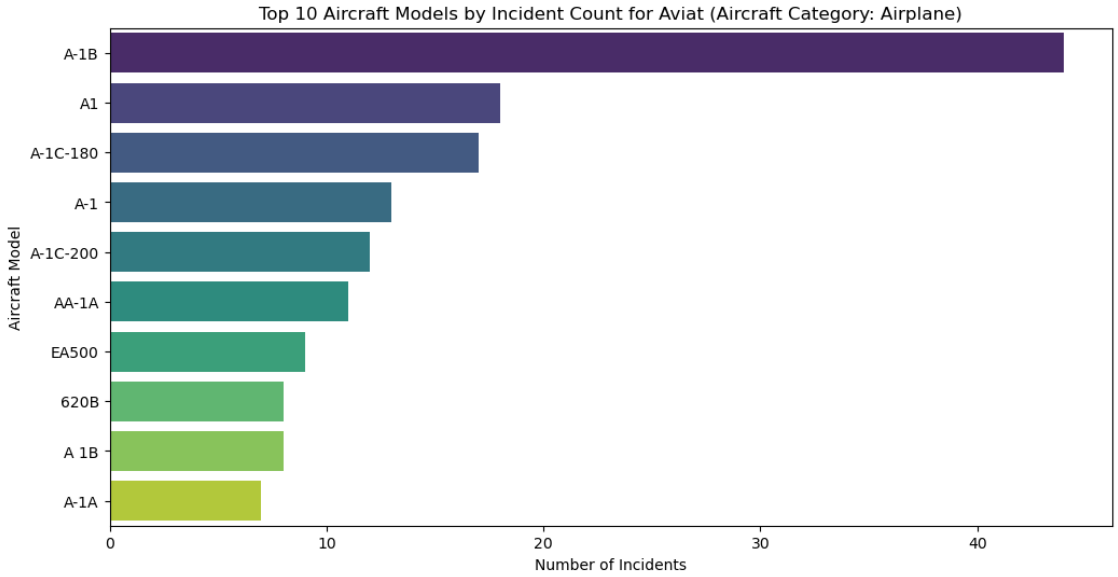

[34]:
```
#  Models by Incident Count Boeing
model_incidents = df_filtered[df_filtered['Make'] == top_make_7]['Model'].
 ↪value_counts().head(10)

plt.figure(figsize=(12, 6))
sns.barplot(x=model_incidents.values, y=model_incidents.index,␣
 ↪hue=make_incidents.index, palette='viridis', legend=False)
plt.title(f'Top 10 Aircraft Models by Incident Count for {top_make_7} (Aircraft␣
 ↪Category: Airplane)')
plt.xlabel('Number of Incidents')
plt.ylabel('Aircraft Model')
plt.show()
```




[35]:
```
#  Models by Incident Count Airbus
model_incidents = df_filtered[df_filtered['Make'] == top_make_8]['Model'].
 ↪value_counts().head(10)

plt.figure(figsize=(12, 6))
sns.barplot(x=model_incidents.values, y=model_incidents.index,␣
 ↪hue=make_incidents.index, palette='viridis', legend=False)
plt.title(f'Top 10 Aircraft Models by Incident Count for {top_make_8} (Aircraft␣
 ↪Category: Airplane)')
plt.xlabel('Number of Incidents')
plt.ylabel('Aircraft Model')
plt.show()
```



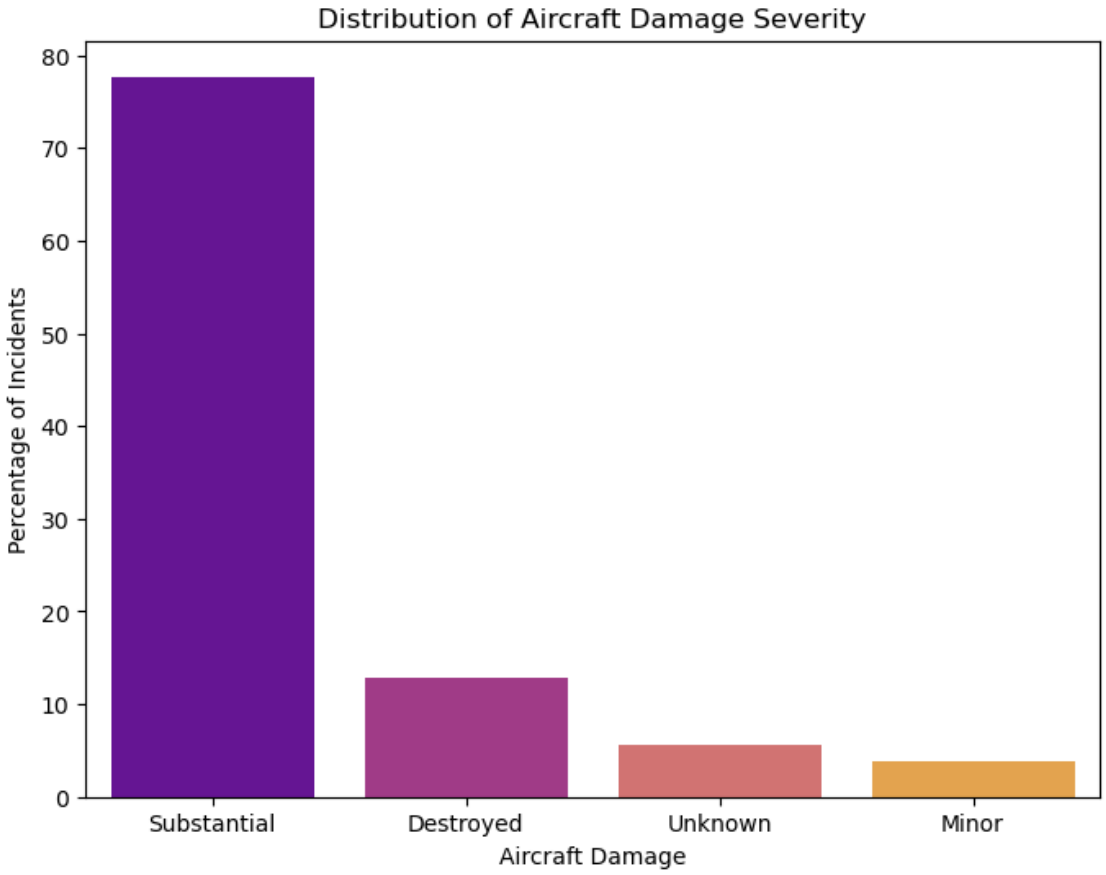

[36]:
```
#  Models by Incident Count Aviat
model_incidents = df_filtered[df_filtered['Make'] == top_make_9]['Model'].
 ↪value_counts().head(10)

plt.figure(figsize=(12, 6))
sns.barplot(x=model_incidents.values, y=model_incidents.index,␣
 ↪hue=make_incidents.index, palette='viridis', legend=False)
plt.title(f'Top 10 Aircraft Models by Incident Count for {top_make_9} (Aircraft␣
 ↪Category: Airplane)')
plt.xlabel('Number of Incidents')
plt.ylabel('Aircraft Model')
plt.show()
```



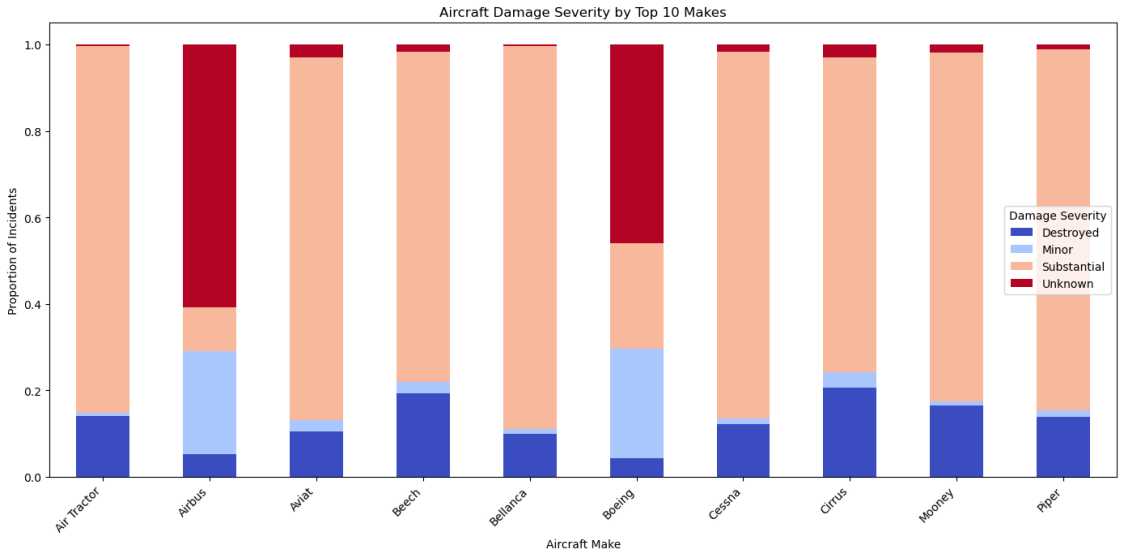

### 1.5.2 Severity of Incidents by Aircraft Damage

- This directly addresses the financial implications of incidents. "Destroyed" or "Substantial" damage means high costs.

[37]:
```
# Distribution of Aircraft Damage in percentage
damage_distribution = df_filtered['Aircraft_damage'].
 ↪value_counts(normalize=True) * 100


plt.figure(figsize=(8, 6))
sns.barplot(x=damage_distribution.index, y=damage_distribution.values,␣
 ↪hue=damage_distribution.index, palette='plasma', legend=False)
plt.title('Distribution of Aircraft Damage Severity')
plt.xlabel('Aircraft Damage')
plt.ylabel('Percentage of Incidents')
plt.show()
```



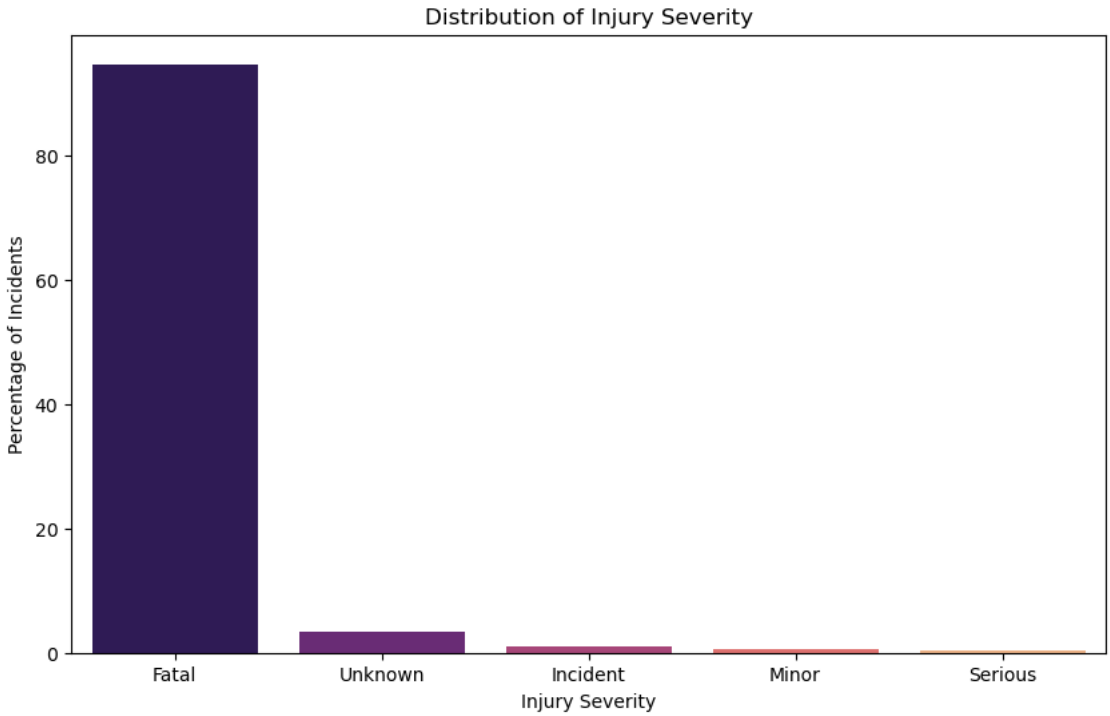

[38]:
```
# Damage severity by Make top 5 makes
top_10_makes = df_filtered['Make'].value_counts().head(10).index
damage_by_make = df_filtered[df_filtered['Make'].isin(top_10_makes)].
 ↪groupby('Make')['Aircraft_damage'].value_counts(normalize=True).
 ↪unstack(fill_value=0)


damage_by_make.plot(kind='bar', stacked=True, figsize=(14, 7), cmap='coolwarm')
plt.title('Aircraft Damage Severity by Top 10 Makes')
plt.xlabel('Aircraft Make')
plt.ylabel('Proportion of Incidents')
plt.xticks(rotation=45, ha='right')
plt.legend(title='Damage Severity')
plt.tight_layout()
plt.show()
```



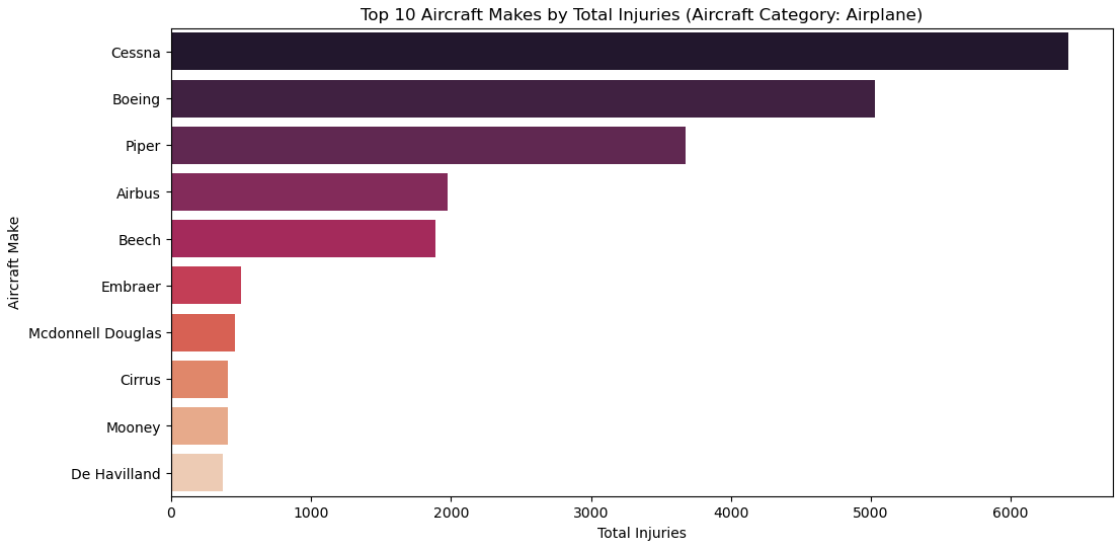

### 1.5.3 Human Impact: Injury Severity

- This is about the human cost. High numbers of fatalities or serious injuries indicate a higher "safety risk" profile for an aircraft type.

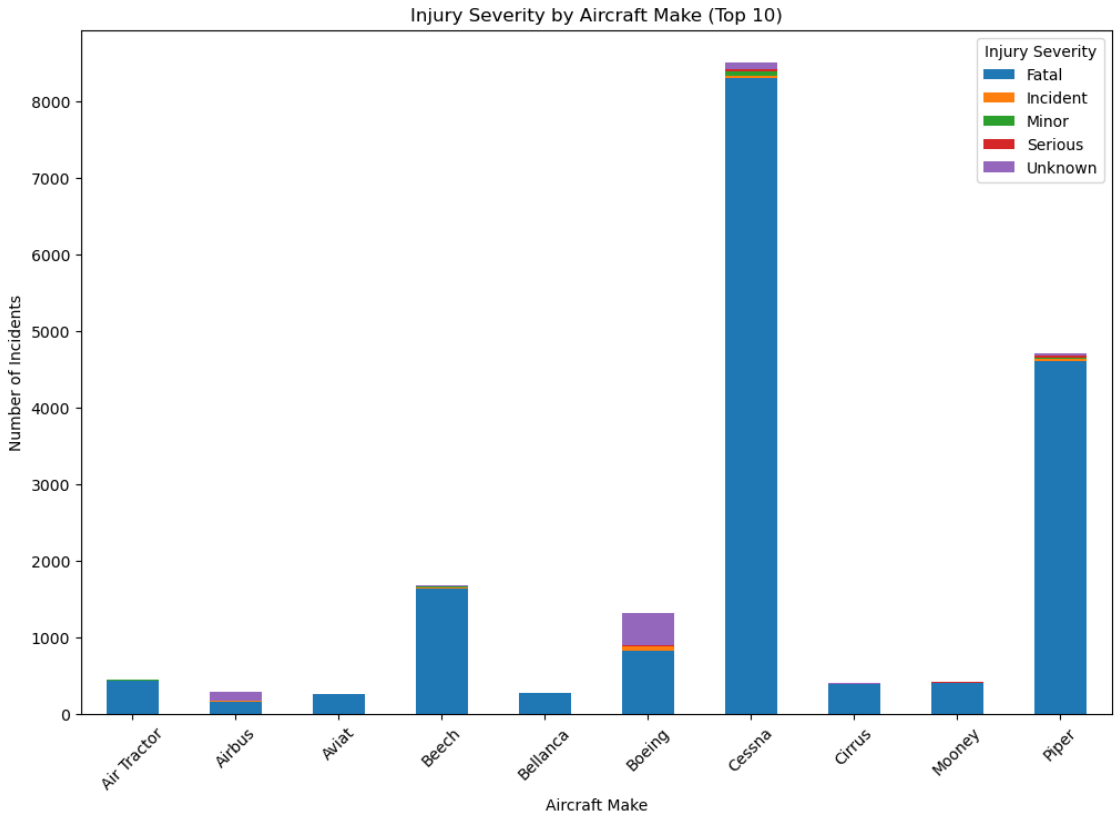
[39]:
```
# Distribution of Injury Severity in percentage
injury_distribution = df_filtered['Injury_Severity'].
 ↪value_counts(normalize=True) * 100

plt.figure(figsize=(10, 6))
sns.barplot(x=injury_distribution.index, y=injury_distribution.values,  hue =␣
 ↪injury_distribution.index, palette='magma', legend=False)
plt.title('Distribution of Injury Severity')
plt.xlabel('Injury Severity')
plt.ylabel('Percentage of Incidents')
plt.show()
```




[40]:
```
# Total Injuries by Make requires aggregate of all injury columns
df_filtered['Total_Injuries'] = df_filtered['Total_Fatal_Injuries'] +␣
 ↪df_filtered['Total_Serious_Injuries'] + df_filtered['Total_Minor_Injuries']

total_injuries_by_make = df_filtered.groupby('Make')['Total_Injuries'].sum().
 ↪nlargest(10)

plt.figure(figsize=(12, 6))
sns.barplot(x=total_injuries_by_make.values, y=total_injuries_by_make.index,␣
 ↪hue=total_injuries_by_make.index, palette='rocket', legend=False)
plt.title('Top 10 Aircraft Makes by Total Injuries (Aircraft Category:␣
 ↪Airplane)')
plt.xlabel('Total Injuries')
plt.ylabel('Aircraft Make')
plt.show()
```



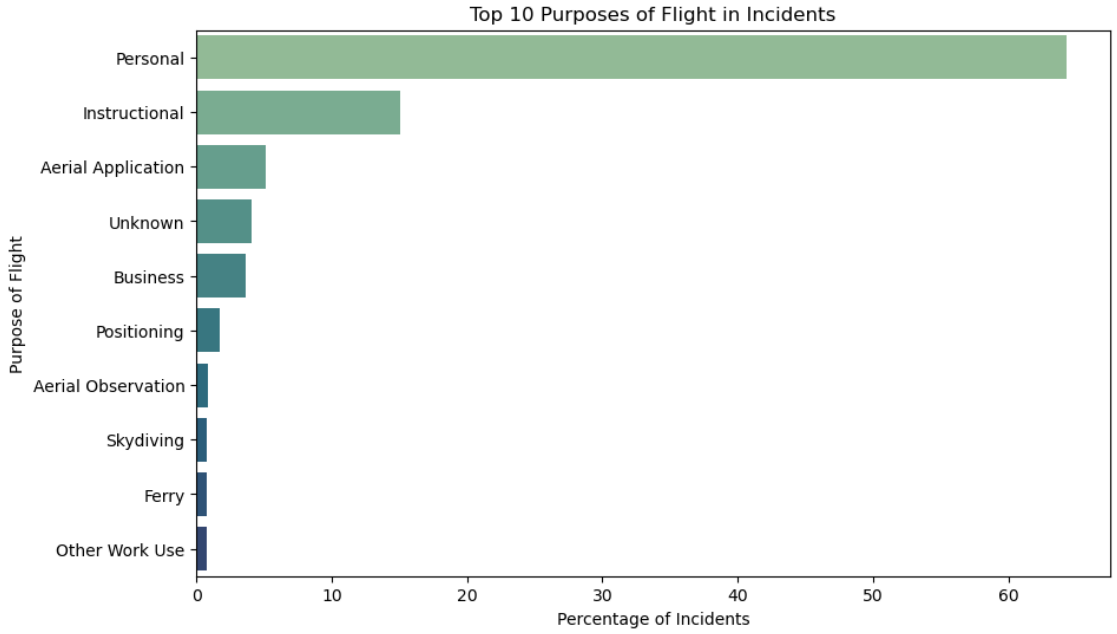

[41]:
```
# Injuries Severity by Make in the top 10
# Get the top 10 makes
top_10_makes = df_filtered['Make'].value_counts().head(10).index

# Filter the DataFrame to include only the top 10 makes
df_top_10_makes = df_filtered[df_filtered['Make'].isin(top_10_makes)]

make_injury = df_top_10_makes.groupby(['Make', 'Injury_Severity']).size().
 ↪unstack(fill_value=0)

# Create the stacked bar chart
make_injury.plot(kind='bar', stacked=True, figsize=(12, 8))

plt.title('Injury Severity by Aircraft Make (Top 10)')
plt.xlabel('Aircraft Make')
plt.ylabel('Number of Incidents')
plt.xticks(rotation=45)
plt.legend(title='Injury Severity')
plt.show()
```



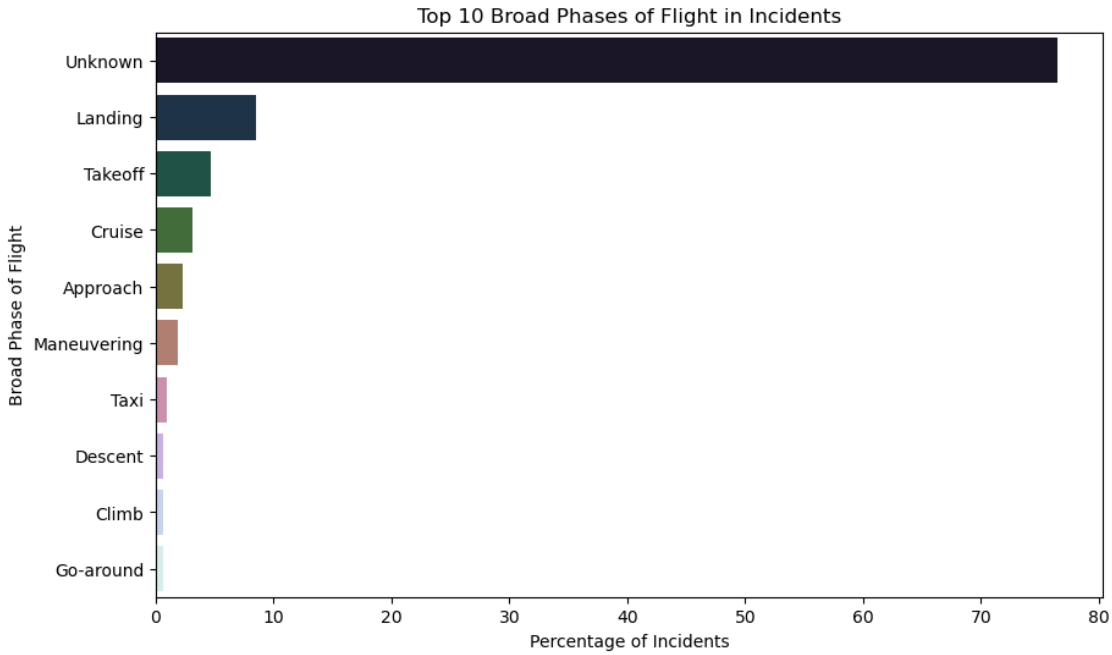

### 1.5.4 Operational Context: Purpose of Flight & Broad Phase of Flight

- Purpose_of_flight, Weather_Condition and Broad_phase_of_flight will give us insights into the circumstances of accidents. For instance, if a specific aircraft model has a high number of landing accidents, it might suggest a design or operational characteristic that makes landing more challenging for that type.

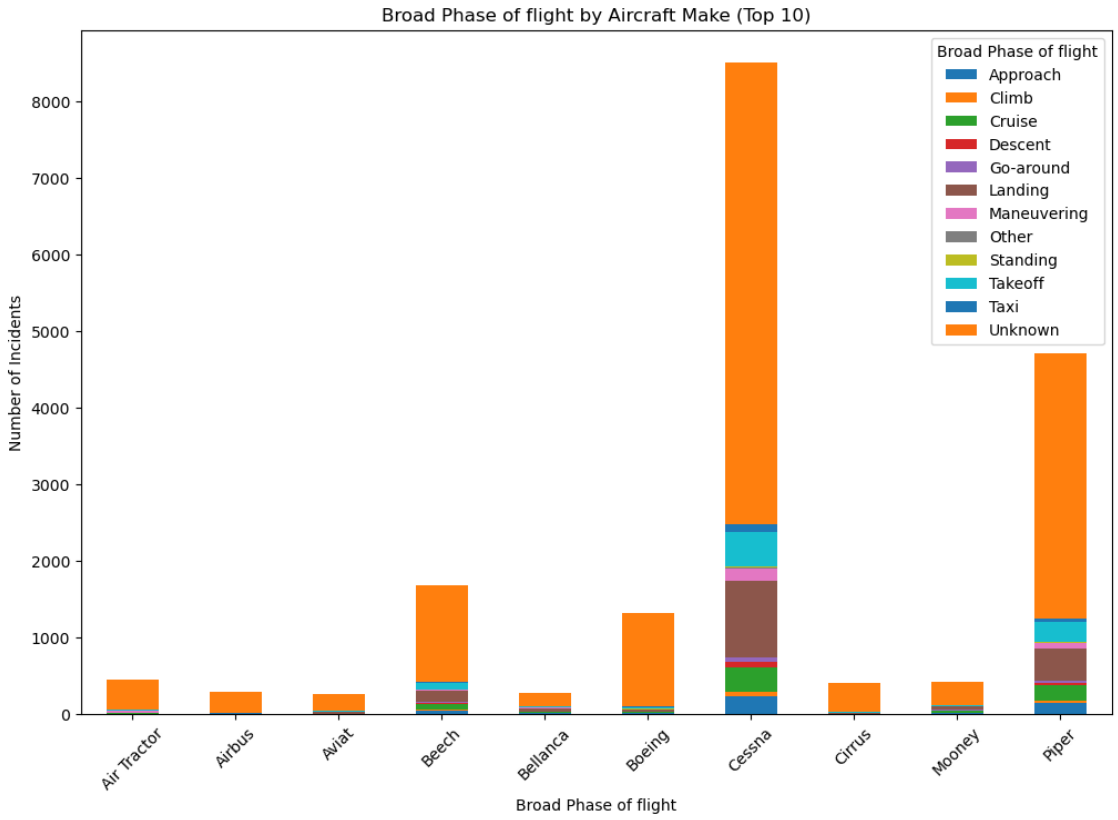
[42]:
```
# Distribution of Purpose of Flight in Incidents
purpose_distribution = df_filtered['Purpose_of_flight'].
 ↪value_counts(normalize=True).head(10) * 100

plt.figure(figsize=(10, 6))
sns.barplot(x=purpose_distribution.values, y=purpose_distribution.index,␣
 ↪hue=purpose_distribution.index, palette='crest', legend=False)
plt.title('Top 10 Purposes of Flight in Incidents')
plt.xlabel('Percentage of Incidents')
plt.ylabel('Purpose of Flight')
plt.show()
```



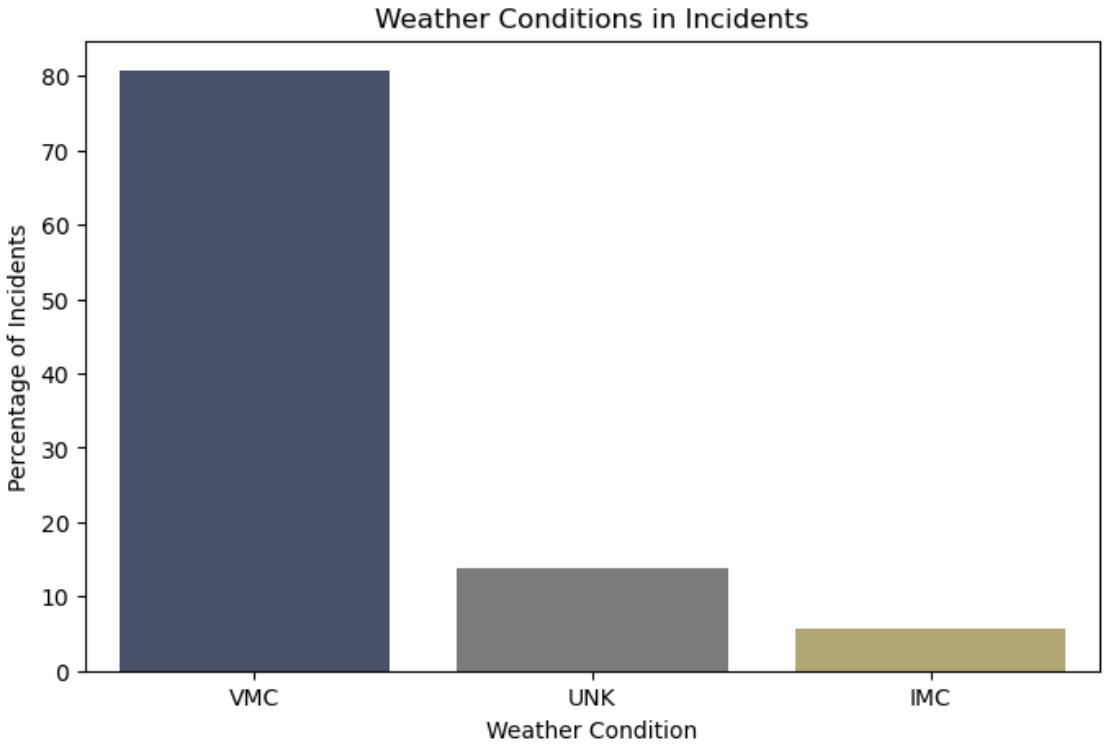

[43]:
```
# Distribution of Broad Phase of Flight in Incidents
phase_distribution = df_filtered['Broad_phase_of_flight'].
 ↪value_counts(normalize=True).head(10) * 100

plt.figure(figsize=(10, 6))
sns.barplot(x=phase_distribution.values, y=phase_distribution.index,␣
 ↪hue=phase_distribution.index, palette='cubehelix', legend = False)
plt.title('Top 10 Broad Phases of Flight in Incidents')
plt.xlabel('Percentage of Incidents')
plt.ylabel('Broad Phase of Flight')
plt.show()
```



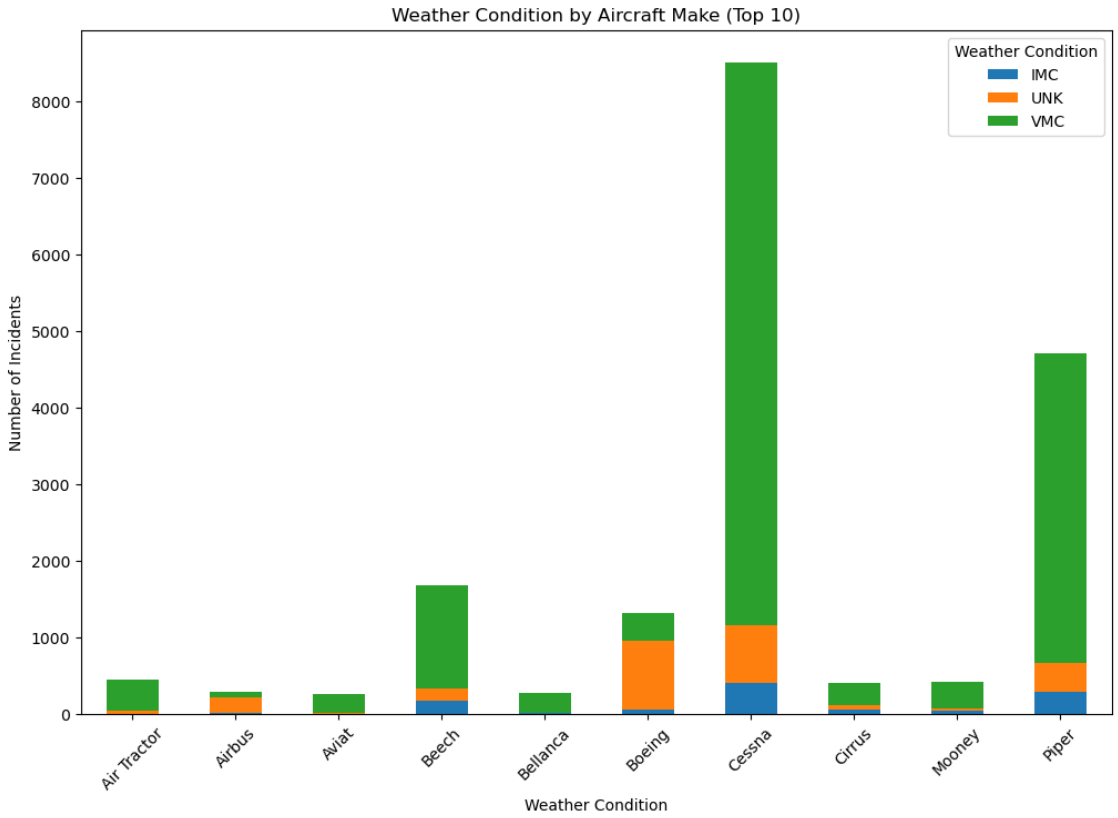

[44]:
```
# Broad Phase of Flight by Make in the top 10
# Get the top 10 makes
top_10_makes = df_filtered['Make'].value_counts().head(10).index

# Filter the DataFrame to include only the top 10 makes
df_top_10_makes = df_filtered[df_filtered['Make'].isin(top_10_makes)]

Broad_Phase = df_top_10_makes.groupby(['Make', 'Broad_phase_of_flight']).size().
 ↪unstack(fill_value=0)

# bar chart
Broad_Phase.plot(kind='bar', stacked=True, figsize=(12, 8))

plt.title('Broad Phase of flight by Aircraft Make (Top 10)')
plt.xlabel('Broad Phase of flight')
plt.ylabel('Number of Incidents')
plt.xticks(rotation=45)
plt.legend(title='Broad Phase of flight')
plt.show()
```




[45]:
```
# Weather Conditions Impact
# Distribution of Weather Conditions in Incidents
weather_distribution = df_filtered['Weather_Condition'].
 ↪value_counts(normalize=True) * 100

plt.figure(figsize=(8, 5))
sns.barplot(x=weather_distribution.index, y=weather_distribution.values,␣
 ↪hue=weather_distribution.index, palette='cividis',legend= False)
plt.title('Weather Conditions in Incidents')
plt.xlabel('Weather Condition')
plt.ylabel('Percentage of Incidents')
plt.show()
```




[46]:
```
# Weather Conditions by Make in the top 10
# Get the top 10 makes
top_10_makes = df_filtered['Make'].value_counts().head(10).index

# Filter the DataFrame to include only the top 10 makes
df_top_10_makes = df_filtered[df_filtered['Make'].isin(top_10_makes)]

Weather_condition = df_top_10_makes.groupby(['Make', 'Weather_Condition']).
 ↪size().unstack(fill_value=0)

# bar chart
Weather_condition.plot(kind='bar', stacked=True, figsize=(12, 8))

plt.title('Weather Condition by Aircraft Make (Top 10)')
plt.xlabel('Weather Condition')
plt.ylabel('Number of Incidents')
plt.xticks(rotation=45)
plt.legend(title='Weather Condition')
plt.show()
```




[47]:
```
df_filtered.to_csv('cleaned_aviation_data.csv', index=False)
```

### 1.6 Summary and Interpretation of Findings

**Based on this dataset- accident data, the primary risk factors associated with aircraft ownership and operation, quantifiable through this dataset, include:**

- **Accident Frequency/Rates:** Certain aircraft Makes (e.g., Cessna, Piper) have a higher number of recorded incidents. While not a true 'rate' without fleet data, higher counts indicate higher exposure to incidents.")
- **Aircraft Damage Severity:** A large proportion of incidents result in 'Substantial' damage, indicating significant financial cost for repairs or replacement.")
- **Human Injury/Fatality Rates:** The occurrence of fatal, serious, and minor injuries is a critical risk factor, both from a human capital and liability perspective.")
- **Operational Phase Vulnerabilities:** Specific phases of flight (e.g., Landing, Takeoff) are significantly more prone to incidents for all aircraft, suggesting these are high-risk operational periods.")
- **Weather Conditions:** While VMC is most common, understanding incidents in IMC (Instrument Meteorological Conditions) or unknown conditions is important for assessing all-weather operational risk.")
- **Purpose of Flight:** Incidents vary by flight purpose but personal appears to be the most common.

*NOTE*: To truly determine 'lowest risk,' we will would need to cross-reference these findings with external data on maintenance costs, fuel efficiency, spare parts availability, and aircraft market value/depreciation.")

## 1.7 Business recommendations

Based on the analysis of the aviation incident dataset which focussed on accident frequency, damage severity, and human impact, here are three concrete business recommendations for the Head of the New Aviation Division:

### 1.7.1 Business Recommendation 1

**Prioritize Aircraft Acquisition of Low-Incidence, Low-Severity Make, e.g *Airbus*, *Bellanca* and *Maule*. Rationale**: The analysis identifies specific aircraft types with the lowest combined risk profiles, demonstrating fewer incidents, less severe damage, and fewer serious injuries. Acquiring these models directly minimizes operational, financial, and safety risks for the new aviation venture.

**Actionable Step**: Immediately commence in-depth financial and operational due diligence, including total cost of ownership and training requirements, exclusively on the identified low-risk aircraft models before broadening the selection.

### 1.7.2 Business Recommendation 2

**Develop and Enforce Rigorous Standard Operating Procedures (SOPs) and Enhanced Pilot Training Curricula Concentrating on *Takeoff* and *Landing* Phases across the entire fleet. Rationale**: The data consistently reveals that Takeoff and Landing phases are universally vulnerable across all aircraft, making proactive risk mitigation in these areas crucial for maximizing safety dividends.

**Actionable Step**: Design and implement mandatory, recurrent simulator training modules for takeoff and landing scenarios, complemented by strict pilot proficiency checks, to enhance safety during these high-risk flight phases.

### 1.7.3 Business Recommendation 3

**Establish a Robust Risk Monitoring Framework that Continuously Tracks Incident Data for the Acquired Fleet and Develops a Contingency Fund for Severe Aircraft Damage. Rationale**: Even for low-risk aircraft, aviation risks are dynamic and can lead to significant financial exposure from substantial damage, thus proactive monitoring and a dedicated contingency fund are essential to address emerging and inevitable high-cost incidents.

**Actionable Step**: Implement a system to regularly review updated incident data for the specific makes and models purchased, identifying any new trends or emerging safety alerts.

[ ]: